

Berger-Hopf Standard Model: A No-Retuning Geometric Reinterpretation Framework for Standard Model Flavor, Couplings, and Generations

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Abstract

The Berger-Hopf Standard Model (BHSM) is a no-retuning geometric reinterpretation framework for Standard Model flavor, couplings, generations, and electroweak-scale structure. This paper packages the frozen BHSM prediction set, claim ledger, falsification ledger, and reproducibility instructions for public GitHub and Zenodo archival. Claims are separated into derived-conditional, verified-test, strong-screen, proxy-audit, open, and forbidden categories. The release does not claim experimental confirmation, community acceptance, QCD confinement, quantum gravity, or a final replacement of the Standard Model.

1 Plain-Language Summary

BHSM treats some Standard Model patterns as allowed notes of a Berger-Hopf internal geometric instrument. The notes are protected or screened internal modes. The project asks whether a fixed, no-retuning geometry can organize flavor hierarchies, coupling screens, generation structure, and no-extra-light-state constraints.

2 Introduction, Motivation, and Scope

The Standard Model is empirically successful but contains many free flavor and Yukawa parameters. BHSM is a conservative research framework that audits whether a Berger-Hopf internal geometry can reinterpret some of those structures. This release supplies frozen outputs, tests, ledgers, and a manuscript package. It does not claim experimental validation, QCD confinement, quantum gravity, or a community-accepted replacement of the Standard Model.

3 Mathematical and Geometric Setup

The canonical geometry and overlap width are frozen:

$$a = \alpha^{-1}/(12\pi^2) = 1.157054135733433, \quad S = 1/(4\pi) = 0.07957747154594767.$$

The Berger scalar spectrum proxy used in the audit is

$$\lambda_{k,j}(a) = a^2(k - 2j)^2 + 2((2j + 1)k - 2j^2),$$

with Hopf charge $q = k - 2j$.

4 Berger-Sphere and Hopf-Fibration Interpretation

The Hopf fibration supplies a fiber/base split. The integer pair (k, j) labels supplied internal modes, while q tracks Hopf charge. The geometry is used as a frozen reinterpretation framework and screen generator, not as a claimed first-principles derivation of the entire Standard Model.

5 Charged-Sector Mode Ledger

Sector	Heavy	Middle	Light
Charged leptons	(0, 0)	(5, 2)	(9, 3)
Up quarks	(0, 0)	(6, 0)	(10, 1)
Down quarks	(0, 0)	(6, 3)	(8, 2)

6 Boundary Operators

The operational/action-linked boundary operators are

$$\Omega_\ell = -q + 2j = 3, \quad \Omega_u = q - 2j = 6, \quad \Omega_d = q + 4j = 12.$$

They are labeled ACTION_LINKED or derived-conditional in this release unless the repository proves a stronger status.

7 Frozen Prediction Set

Quantity	BHSM_BARE_V1	BHSM_DRESSED_V1_CANDIDATE	Changed
c/t	0.008310500554068288	0.004155250277034144	True
u/t	1.2690463017606151e-05	1.2690463017606151e-05	False
s/b	0.021933971495439474	0.021933971495439474	False
d/b	0.0011165200546001757	0.0011165200546001757	False
sin_theta_13	0.0035623676140463315	0.0035623676140463315	False

8 Bare and Dressed Prediction Branches

BHSM_BARE_V1 is the alpha-anchored Berger-Hopf overlap model. BHSM_DRESSED_V1_CANDIDATE applies $Z_{\text{virt}}^{u,2} = 1/2$ only to c/t . It leaves u/t , CKM $\sin \theta_{13}$, down-sector ratios, lepton ratios, gauge outputs, Higgs/electroweak outputs, H_T rows, and scalar rows unchanged.

9 No-Retuning Rule

Any post-freeze adjustment of a , S , modes, tolerance bands, outputs, or Z_{virt} based on residuals invalidates the frozen prediction set.

10 Coupling Outputs and Matching-Scale Caveats

Quantity	BHSM output	Status
alpha_1	0.01688686394038963	matching screen
alpha_2	0.03377372788077926	matching screen
alpha_3	0.1182080475827274	matching screen
sin2_theta_w	0.23076923076923078	matching screen
alpha_em_inv_mew	128.30485721416164	matching screen

Full higher-loop and threshold RG matching remains open/future work.

11 Mass-Ratio and CKM Tables

Mass-ratio and CKM outputs are frozen screens; quark mass rows remain scheme-sensitive. CKM screen values include $\sin\theta_{12} = 0.2256184580048353$, $\sin\theta_{23} = 0.04386794299087895$, $\sin\theta_{13} = 0.0035623676140463315$, $\delta_{\text{CP}} = 1.1283791670955126$, and $J_{\text{CKM}} = 3.1011702945437805 \times 10^{-5}$.

12 Higgs, H_T Gap, and Scalar-Decoupling Audit

The Higgs/electroweak row is a strong numerical screen. H_T no-extra-light-state status remains PROXY_AUDIT/open full spectrum in this release. Scalar/topographic decoupling remains a finite-basis scaffold/open full action proof.

13 Tests and Reproducibility

Run:

```
python -m pip install -e .  
python -m pytest
```

The release also includes `tests/test_final_paper_release_package.py`.

14 Claim-Status Ledger

Claims are categorized as DERIVED_CONDITIONAL, VERIFIED_TEST, STRONG_SCREEN, PROXY_AUDIT, OPEN, and FORBIDDEN. See `docs/claim_status_table.md`.

15 Forbidden Claims

This release does not claim a full derivation of the Standard Model from first principles, a proof of confinement, a proof of quantum gravity, a final replacement of the Standard Model, experimental confirmation by the particle-physics community, or post-data retuning.

16 Limitations, Open Problems, and Falsification Criteria

Open work includes the full twisted Dirac/ H_T spectrum beyond proxy/scaffold audits, scalar/topographic decoupling beyond scaffold status, precision QCD/RG threshold matching, independent community review, and external empirical validation. Falsification criteria F1–F9 are summarized in `docs/falsification_criteria.md`.

17 Citation and Archival Information

Use `CITATION.cff`. Zenodo archival should be triggered by a GitHub release if integration is enabled. Do not manually invent a DOI.

18 AI-Assisted Drafting Acknowledgment

AI-assisted drafting/coding support was used to organize repository artifacts, tests, manuscript text, and release metadata. Scientific responsibility, claim discipline, and release approval remain with the author.